



US Army Corps
of Engineers®

Mississippi River at Head of Passes

New Orleans District
Last Updated 9/11/2025

Gage Datasheet

Name Mississippi River at Head of Passes

Gage Type DCP

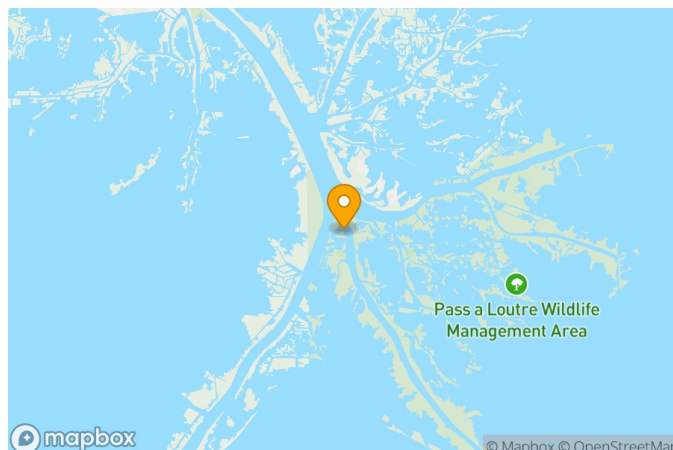
Maintained By U.S. Army Corps of Engineers,
New Orleans District (CEMVN)

USACE Code 01545

Units of Measure FEET

Current QA Complete Date 9/11/2025

Historic QA Complete Date 9/11/2025



Latitude: 29.13377778 **Longitude:** -89.2465

Description

First historic record found 12/23/19422. Gage originally set in MSL = 0'. Gage on the Mississippi River near Head of Passes.

Notes

Originally installed just downstream on the right descending bank in South Pass, moved to the Head of Passes Light at the junction of South Pass and Southwest Pass in 09/30/2020 and moved back to original location 07/27/2022 after dredging completed and access to the original site cleared up

Gage Calibrations

Vertical Datum	Zero Elevation	Period	Calibration Date	Calibration Divergence Date	Calibration Notes
NAVD88 (2009.55)			6/30/2015		gage divergence determined by TBM Head A, elevation = 2.41'; gage zero = -0.15' NAVD88 (2009.55) from 2018 survey; gage surveyed again on 7/29/22 using PBM Pilot Ecc (5.23' NAVD88 (2009.55)) and only found gage zero = -0.018'.
NAVD88 (2004.65)			1/15/2008		Gage inspection from 1/15/08 showed new gage was set to the same water surface elevation as the existing gage; unable to determine datum adjustment from NAVD88 to NGVD29. Gages were surveyed using TBM Head 080806 elevation = 4.39'. No close out survey for this gage found before new staff installed on 02/26/2010
GAGE DATUM [formerly NGVD29]			2/28/1994	1/15/2008	Gage surveyed using TBM Square elevation 5.10' Missing survey documentation between 02/28/1994 and 01/15/2008. Unable to determine when gage was reset to NAVD88

GAGE DATUM [formerly NGVD29]			1/5/1984	2/28/1994	Gage surveyed and reset using TBM Head elevation 2.71' Gage resurveyed 10/26/1984 using PBM Head 3 elevation = 4.54'. No adjustment needed. Gage resurveyed 10/24/1985 using PBM Head 3 elevation = 4.54'. No adjustment needed. Gage resurveyed 05/27/1989 using PBM Willow elevation = 3.82'. No adjustment needed. Gage resurveyed 05/30/1991 using PBM Head 3 elevation = 4.54'. No adjustment needed. Gage resurveyed 05/01/1992 using TBM Square elevation = 5.10'. No adjustment needed. Gage resurveyed 03/09/1993 using TBM Square elevation = 5.10'. No adjustment needed. Gage resurveyed 02/28/1994 using TBM Square elevation = 5.10'. No adjustment needed.
GAGE DATUM [formerly NGVD29]			7/20/1964	1/5/1984	Gage surveyed and set using PBM Mi 0.25 elevation = 5.21' Gage surveyed and reset 06/15/1966 using PBM Mi 0.25 elevation = 5.21' Gage surveyed 11/14/1966 using TBM Fire Ring elevation = 6.18' Gage surveyed 03/06/1968 using PBM Mi 0.25 elevation = 5.21' Gage surveyed 04/06/1970 using PBM Mi 0.25 elevation = 5.21' Gage surveyed 05/26/1971 using TBM Cement Pile elevation = 5.84' Gage surveyed 12/29/1972 using TBM Cement Pile elevation = 5.84' Gage surveyed and reset 01/05/1984 using PBM Head 3 RM 7 elevation = 2.71'
GAGE DATUM [formerly NGVD29]			10/8/1959	7/20/1964	Gage surveyed and set using Bolt TP elevation 6.80'
GAGE DATUM [formerly NGVD29]			12/17/1957	10/8/1959	Gage surveyed and reset 12/17/1957 using PBM "Wise" elevation 4.39' Gage surveyed and reset to same 08/19/1958 using TBM Bolt elevation = 6.80' Gage surveyed and reset to same 10/06/1959 using PBM Sun elevation = 4.96' Gage surveyed and reset to same 10/08/1959 using Bolt TP elevation = 6.80'
GAGE DATUM [formerly NGVD29]			5/15/1956	12/12/1957	Gage surveyed and reset using PBM "Wise" elevation 4.39'.
GAGE DATUM [formerly NGVD29]			1/15/1954	5/15/1956	Gage surveyed and reset using TBM Willow elevation 5.022'
GAGE DATUM [formerly NGVD29]			10/22/1952	1/15/1954	Gage surveyed and reset using BM BURR 1950 elevation 7.940'
GAGE DATUM [formerly NGVD29]			12/12/1946	10/22/1952	Gage surveyed and reset 10/22/1952 using TBM Willow elevation = 6.404' Gage surveyed 07/27/1950 using BM Willow (unrelated to previous) elevation = 5.083' Gage surveyed and reset 10/22/1952 using BM Burr 1950 elevation = 7.940'
GAGE DATUM [formerly NAVD88]			10/2/1944	12/12/1946	Gage reset to NGVD29 10/02/1944 using TP H of P elevation = 7.541'
GAGE DATUM [formerly NGVD29]			12/6/1943	10/2/1944	First instance of Gage no 21 on 12/09/1943. Gage set using T.P. Head of SWP elevation 4.572'. Gage 0' set to MSL 0'.

Gage Calibration Adjustments

Calibration Vertical Datum	Target Vertical Datum	Adjustment Factor	Adjustment Start Date	Adjustment End Date	Adjustment Notes
Target Epoch:					
NAVD88 (2009.55)	MLG	3.03	5/26/2020		Gage adjustment value is based on 3.5' calibration value to MLLW, as defined by CEPD report dated 7/10/2015. Gage adjustment value is also based on the -0.15' gage divergence as measured on 9/30/2018. This adjustment value is relative to MLLW (2012-2016).

Calibration Vertical Datum	Target Vertical Datum	Adjustment Factor	Adjustment Start Date	Adjustment End Date	Adjustment Notes
NAVD88 (2009.55)	MLG	3.17	10/1/2018	5/25/2020	Gage adjustment value is based on 3.5' calibration value to MLLW, as defined by CEPD report dated 7/10/2015. Gage adjustment value is also based on the -0.15' gage divergence as measured on 9/30/2018. This adjustment value is relative to MLLW (2007-2011).
Target Epoch: OTHER					
NAVD88 (2009.55)	MLLW	-0.47	5/26/2020		Gage adjustment value is based on MLLW (2012-2016) defined in EDR-OD-01, Revision 2 dated 5/26/2020. Gage adjustment value is also based on the -0.15' gage divergence as measured on 9/30/2018.
NAVD88 (2009.55)	MLG	3.32	7/10/2015	9/30/2018	Adjustment value is based on 3.5' calibration value to MLLW, as defined by CEPD report dated 7/10/2015. This current adjustment is relative to MLLW (2007-2011).
NAVD88 (2004.65)	NAVD88	0.19	2/27/2010		Add 0.19' to gage data to adjust back to NAVD88 2004.65. Surveyed conducted 08/16/2010 using TBM Head B elevation = 13.66'
NAVD88 (2004.65)	NAVD88	0.14	1/15/2008	2/26/2010	Add 0.14' to gage data to adjust back to NAVD88 2004.65. Survey conducted 02/26/2010 using TBM Bolt elevation = 4.48'
NGVD29	NGVD29	0	5/30/1991	5/31/1991	No adjustment needed, previous benchmarks destroyed and unusable. New benchmarks set using the staff as a reference point.
NGVD29	NGVD29	-0.174	10/22/1952	1/15/1954	Add -0.174' to gage data to adjust to NGVD29. Gage surveyed and reset using TBM Willow elevation = 5.022'
NGVD29	NGVD29	-0.12	12/12/1946	10/22/1952	Add -0.12' to gage data to adjust to NGVD29. Gage surveyed and reset 10/22/1952 using BM Burr 1950 elevation = 7.940'
NAVD88	NGVD29	-0.316	10/2/1944	12/12/1946	Add -0.316' to gage data to adjust back to NGVD29. Gage surveyed and reset 12/12/1946 using TBM Willow elevation = 6.404'
NGVD29	NGVD29	-0.1	12/6/1943	10/2/1944	Add -0.10' to gage data to adjust to NGVD29. Staff surveyed and set 10/02/1944 using TP HofP elevation = 7.541'.
Target Epoch: OPUS					
NAVD88 (2004.65)	NAVD88	-0.217	1/15/2008		Add -0.217' to gage data to adjust to NAVD88 OPUS 2021. Survey conducted 07/29/2022 using PBM Pilot ECC OPUS shared solution elevation = 5.031'
Target Epoch: NO EPOCH					
NGVD29	NGVD29	-0.43	12/29/1972	1/5/1984	Add -0.43' to gage data to adjust to NGVD29
NGVD29	NGVD29	0.06	2/28/1964	7/20/1964	Add 0.06' to gage data to adjust to NGVD29. Gage surveyed and set using PBM Mi 0.25 elevation = 5.21'
NGVD29	NGVD29	0.03	10/8/1959	2/27/1964	Add 0.03' to gage data to adjust to NGVD29.
NGVD29	NGVD29	-0.21	10/6/1959	10/8/1959	Add -0.21' to gage data to adjust to NGVD29. Gage surveyed and reset 10/08/1959 using TP Bolt elevation = 6.80'
NGVD29	NGVD29	-0.28	5/15/1956	12/17/1957	Add -0.28' to data to adjust to NGVD29. Surveyed and reset using PBM Wise (1956) elevation = 4.39'
NGVD29	NGVD29	-0.14	5/15/1956	12/12/1957	Add -0.14 to gage data to get to NGVD29. Gage surveyed and reset 12/12/1957 using PBM Wise elevation = 4.39'
Target Epoch: 2009.55					
NAVD88 (2004.65)	NAVD88	-0.018	10/1/2018	7/27/2022	Add -0.018' to gage data to adjust to NAVD88 2009.55. RTK Survey conducted 07/27/2022 using PBM Pilot ECC elevation = 5.230'.
NAVD88 (2004.65)	NAVD88	-0.15	8/14/2014	9/30/2018	Add -0.15' to gage data to adjust to NAVD88 2009.55. Survey conducted using TMB Head A elevation 2.41'
NAVD88 (2004.65)	NAVD88	-1.06	10/18/2013	8/13/2014	Add -1.06' to gage data to adjust to NAVD88 2009.55. Survey conducted using TBM Pal elevation = 2.52'
NAVD88 (2004.65)	NAVD88	-1.12	1/15/2008	10/17/2013	Add -1.12' to gage data to adjust to NAVD88 2009.55. Surveyed 10/17/2013 using TBM Head A elevation = 2.528'.
Target Epoch: 2007-2011					

Calibration Vertical Datum	Target Vertical Datum	Adjustment Factor	Adjustment Start Date	Adjustment End Date	Adjustment Notes
NAVD88 (2009.55)	MLLW	-0.33	10/1/2018	5/25/2020	Gage adjustment value is based on MLLW defined in CEPD report dated 7/10/2015, and is relative to MLLW (2007-2011). Gage adjustment value is also based on the -0.15' gage divergence as measured on 9/30/2018.
NAVD88 (2009.55)	MLLW	-0.18	7/10/2015	9/30/2018	Adjustment value is defined in CEPD report dated 7/10/2015, and is relative to 2007-2011 tidal epoch.
Target Epoch: 2004.65					
NGVD29	NAVD88	0	2/28/1994	1/15/2008	Gage surveyed in NAVD88 2004.65 using BM Head 080806 and Gage 0' NGVD29 = 0' NAVD88 2004.65

Gage Statuses

Status Date	Outside Reference Stage (Feet)	Sensor Stage (Feet)	Sensor Offset Elevation (Feet)	Recorder Stage (Feet)	Precipitation (Inches)	Discharge Calculations Performed	Battery Replacement Performed	Range Maintenance Performed	Equipment Swap Performed	Calibration Survey Performed	Adjustment Survey Performed	Notes
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Gage Equipment

Equipment Type	Equipment Subtype	Model Number	UPC Number	Serial Number	HIF Number	Firmware Version	Installed Date	Removed Date	Remarks
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